where $m_{ij} = m_i m_j/(m_i + m_j)$ is the "reduced mass", $v_{ij} = \hat{\boldsymbol{q}}_{ij} \cdot \boldsymbol{v}_{ij}$ and $F^{\mathrm{C}}_{ij}(r_{ij})$ is the magnitude of the conservative force (8.82). This equation is an Ornstein-Uhlenbeck process with the exact (in the sense of distributions) solution [200]

$$v_{ij}(t) = \frac{F^{\mathrm{C}}_{ij}}{\tau m_{ij}} + e^{-\tau t}\left(v_{ij}(0) - \frac{F^{\mathrm{C}}_{ij}}{\tau m_{ij}}\right) + \sqrt{\frac{\mathrm{k_B T}(1-e^{-2\tau t})}{m_{ij}}}\mathrm{R}_{ij}(t), \tag{8.104}$$

for $i < j$ with $\tau = \gamma\omega^{\mathrm{D}}/m_{ij}$, where $v_{ij}(0)$ are the initial relative velocities and $\mathrm{R}_{ij}(t)$ are normally distributed variables with zero mean and unit variance. Thus the incremental velocities are given by

$$v_{ij} = v_{ij}(t) - v_{ij}(0) = \left(v_{ij}(0) - \frac{F^{\mathrm{C}}_{ij}}{\tau m_{ij}}\right)(e^{-\tau t} - 1) + \sqrt{\frac{\mathrm{k_B T}(1-e^{-2\tau t})}{m_{ij}}}\mathrm{R}_{ij}(t). \tag{8.105}$$

The momenta are then obtained from

$$\boldsymbol{p}_i^{n+1} = \boldsymbol{p}_i^n + m_{ij} v_{ij} \hat{\boldsymbol{q}}_{ij}^n, \tag{8.106}$$

$$\boldsymbol{p}_j^{n+1} = \boldsymbol{p}_j^n - m_{ij} v_{ij} \hat{\boldsymbol{q}}_{ij}^n, \tag{8.107}$$

thus defining the propagator $e^{h\mathcal{L}_{\mathrm{S}_{i,j}}}$ for each particle pair. The DPD-Trotter propagator is then

$$\exp\left(h\hat{\mathcal{L}}_{\text{DPD-Trotter}}\right) = \exp\left(\frac{h}{2}\mathcal{L}_{\mathrm{S}}\right)\exp\left(h\mathcal{L}_{\mathrm{A}}\right)\exp\left(\frac{h}{2}\mathcal{L}_{\mathrm{S}}\right),$$

where the momentum part is defined by

$$\exp\left(\frac{h}{2}\hat{\mathcal{L}}_{\mathrm{S}}\right) = \exp\left(\frac{h}{2}\mathcal{L}_{\mathrm{S}_{N-1,N}}\right)\cdots\exp\left(\frac{h}{2}\mathcal{L}_{\mathrm{S}_{1,3}}\right)\exp\left(\frac{h}{2}\mathcal{L}_{\mathrm{S}_{1,2}}\right).$$

8.9.3 *Extended Variable Momentum-Conserving Thermostats: Pairwise Nosé-Hoover/Nosé-Hoover-Langevin Methods*

Because of the form of the force laws, the number of interaction terms (and computational cost) involved in the three force terms in DPD are roughly balanced. In this setting the computational time needed to generate the independent random number streams can be significant (depending on the particle density and cutoff radius), particularly in very large systems solved in parallel where communication between cells can require a significant amount of time [1, 301]. For this reason, Allen and Schmid [8] proposed a Nosé-Hoover type thermostatting scheme in